

FREEDOM INTERNATIONAL SCHOOL

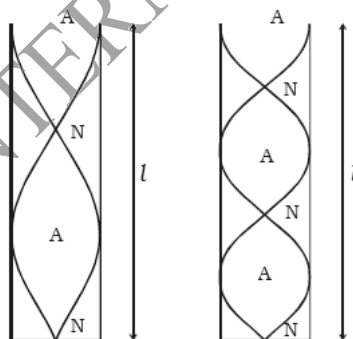
WORKSHEET

PHYSICS

WAVES

CLASS XI

1. Sound is simultaneously produced at the ends of the two strings of the same length, one of rubber and the other of steel. In which string will the sound reach the other end earlier and why?
2. Does the sound of a bomb explosion travel faster than the sound produced by a humming bee?
3. How does the frequency of a vibrating wire change, when the attached load is immersed in water?
4. An incident wave is represented by $y(x, t) = 20 \sin(2x - 4t)$. Write the expression for reflected wave:
 - (i) from a rigid boundary
 - (ii) from an open boundary
5. The figure shows two vibrating modes of an air column. Find the ratio of frequencies of the two modes.



6. An organ pipe is in resonance with a tuning fork. What change will have to be done in the length L to maintain the resonance, if (i) the temperature increases, (ii) hydrogen is filled in place of air and (iii) pressure becomes higher?
7. Tube A has both ends open, while B has one end closed. Otherwise the two tubes are identical. What is the ratio of fundamental frequency of the tubes A and B?
8. The length of a sonometer wire is 0.75m and density $9 \times 10^3 \text{ kg m}^{-3}$. It can bear a stress of $8.1 \times 10^8 \text{ N/m}^2$ without exceeding the elastic limit. What is the fundamental frequency that can be produced in the wire?

9. A stone dropped from the top of a tower 300 m high splashes into the water of a pond near the base of the tower. When is the splash heard at the top? Speed of sound in air = 340 m/s; $g = 9.8 \text{ m/s}^2$.
10. A transverse harmonic wave on a string is described by $y(x, t) = 3.0 \sin(36t + 0.018x + \pi/4)$, where x, y are in cm and t in s. The positive direction of x is from left to right.
- (i) Is this a travelling or stationary wave? If it is travelling, what are the speed and direction of its propagation?
 - (ii) What are its amplitude and frequency?
 - (iii) What is the initial phase at the origin?
 - (iv) What is the least distance between two successive crests in the wave?

FREEDOM INTERNATIONAL SCHOOL